University of Stuttgart
Collaborative
Artificial Intelligence

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Machine Perception and Learning

for Collaborative Intelligent Systems

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Multimodal Learning

- Motivation of multimodal learning
- Multimodal Representation Learning
- Multimodal Alignment
- Multimodal Reasoning

Interactive Machine Learning

- Introduction
- Semi-supervised Learning
- Active Learning
- Other Interesting AL Techniques used in Practice
- Graph-based Active and Semi-supervised Methods
- Deep Active Learning

Interactive Machine Learning

Introduction

- Andreas Holzinger (University of Natural Resources and Life Sciences, Vienna, Austria / TU Graz)
- Maria-Florina Balcan (CMU)
- Jacob Gildenblat: Overview of Active Learning for Deep Learning

See *https://jacobgil.github.io/deeplearning/activelearning#active-learning-for-deep-learning*

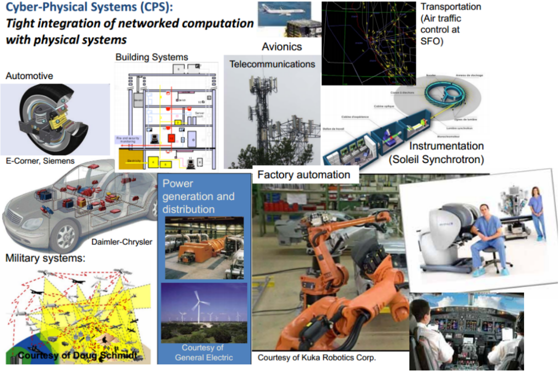
- Most machine learning (ML) applications today use so-called **automatic** machine learning (aML) approaches
- automatic ML = algorithms that interact with agents and can optimize their learning behaviour through this interaction

Source: Andreas Holzinger, University of Natural Resources and Life Sciences, Vienna

Introduction – Autonomous Vehicles ("Google Car")

Source: Andreas Holzinger, University of Natural Resources and Life Sciences, Vienna

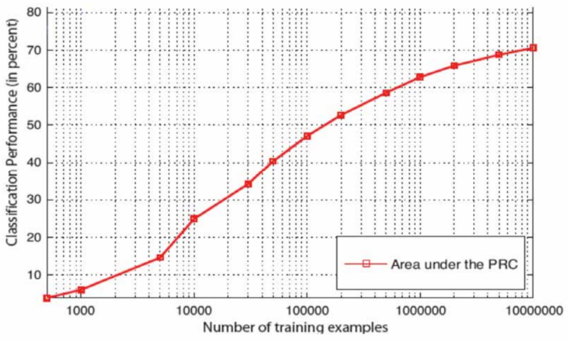
Slide source: [Dietterich and Horvitz, 2015]



Source: Andreas Holzinger, University of Natural Resources and Life Sciences, Vienna

Slide source: [Seshia et al., 2015]

Large datasets are necessary!



Source: Andreas Holzinger, University of Natural Resources and Life Sciences, Vienna

Slide source: [Sonnenburg et al., 2006]

Introduction – Does this work across applications?

Source: Andreas Holzinger, University of Natural Resources and Life Sciences, Vienna

Medical decision making as a search task in $\mathcal{H}$
Problem: Time (t)

Search in an arbitrarily high-dimensional space < 5 minutes!

Source: Andreas Holzinger, University of Natural Resources and Life Sciences, Vienna

- Maximising Expected Utility Theory

- Sometimes we do not have “big data”, where aML algorithms benefit
- We might have:
    - Small or multiple datasets
    - Rare events - how do we get training examples?
    - NP-hard problems:
        - Subspace clustering
        - Protein-Folding
        - k-Anonymisation
        - Graph Colouring, Category Discovery, etc.

- Sometimes we (still) need a human in the loop

- interactive Machine Learning (iML) := algorithms that interact with agents* and that can optimise their learning behaviour through this interaction
- * where the agents can be humans!

Slide source: [Holzinger, 2015]

Introduction – A group of experts in the loop

Source: Andreas Holzinger, University of Natural Resources and Life Sciences, Vienna

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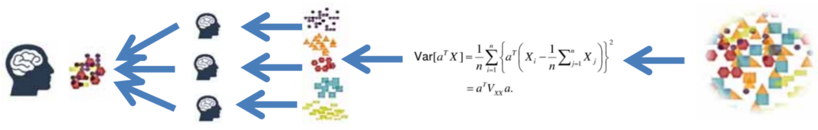
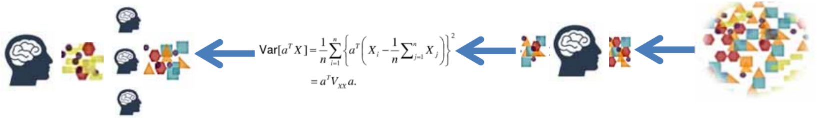
Source: Andreas Holzinger, University of Natural Resources and Life Sciences, Vienna

- Unsupervised ML: The algorithm is applied on the raw data and learns fully automatic
- Humans can check the results at the end of the pipeline



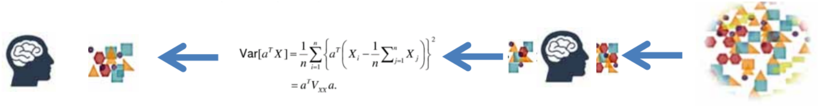
Source: Andreas Holzinger, University of Natural Resources and Life Sciences, Vienna

- Supervised ML: Humans are providing labels for the training data and/or select features to feed the algorithm to learn
- The more samples the better
- Humans can check results at the end of the pipeline



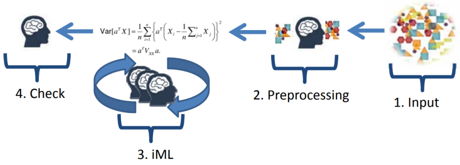
Source: Andreas Holzinger, University of Natural Resources and Life Sciences, Vienna

- Semi-supervised ML: A mixture of A and B – mixing labelled and unlabelled data so that the algorithm can find labels according to a similarity measure to one of the given groups



Source: Andreas Holzinger, University of Natural Resources and Life Sciences, Vienna

- The human is seen as an agent involved in the actual learning phase, influencing measures such as distance or cost functions step by step



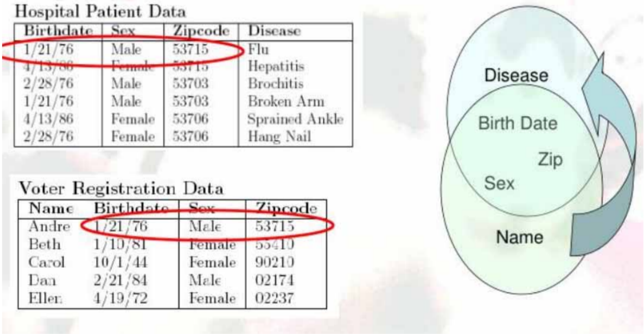
Source: Andreas Holzinger, University of Natural Resources and Life Sciences, Vienna

- Constraints of humans: robustness, subjectivity, transfer?
- Open questions: evaluation, reproducibility

Slide source: [Holzinger, 2016]

- k-Anonymity
- Protein Folding
- Subspace Clustering

87% of the population in the USA can be uniquely reidentified by zip-code, gender and date of birth



Hospital Patient Data

Birthdate	Sex	Zipcode	Disease
1/21/76	Male	53715	Flu
4/13/86	Female	53715	Hepatitis
2/28/76	Male	53703	Brochitis
1/21/76	Male	53703	Broken Arm
4/13/86	Female	53706	Sprained Ankle
2/28/76	Female	53706	Hang Nail

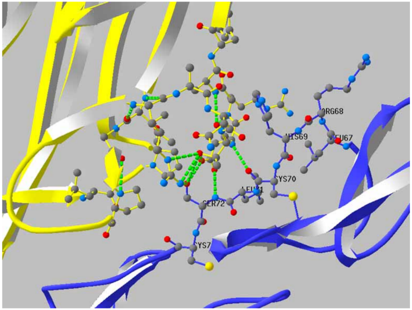
Voter Registration Data

Name	Birthdate	Sex	Zipcode
Andre	1/21/76	Male	53715
Beth	1/10/81	Female	55410
Carol	10/1/44	Female	90210
Dan	2/21/84	Male	02174
Ellen	4/19/72	Female	02237

Source: Andreas Holzinger, University of Natural Resources and Life Sciences, Vienna

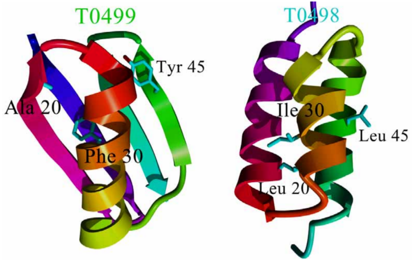
Slide source: [Sweeney, 2002]

Proteins are the building block of life



Source: Andreas Holzinger, University of Natural Resources and Life Sciences, Vienna

Slide source: [Wiltgen et al., 2007]

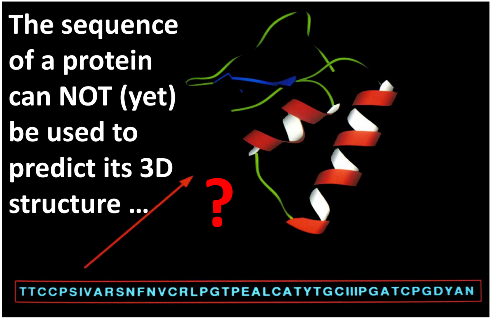


Source: Andreas Holzinger, University of Natural Resources and Life Sciences, Vienna

Source: Andreas Holzinger, University of Natural Resources and Life Sciences, Vienna

Slide source: [He et al., 2008]

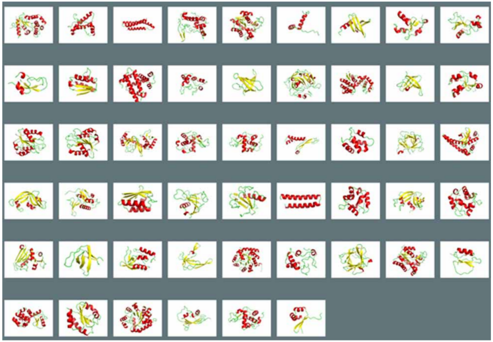
Protein prediction is an old – still unsolved - problem



Source: Andreas Holzinger, University of Natural Resources and Life Sciences, Vienna

Slide source: [Anfinsen, 1973]

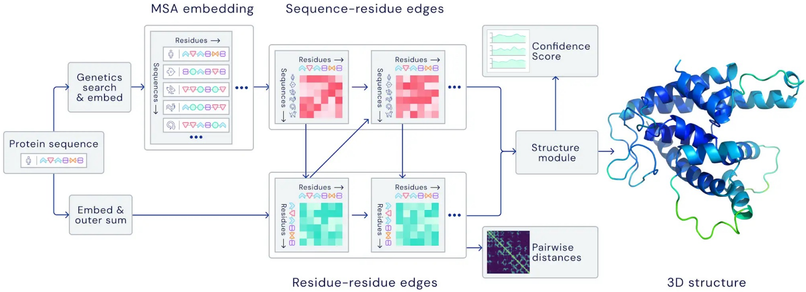
2015: Automatic ML methods do not work well enough*



Source: Andreas Holzinger, University of Natural Resources and Life Sciences, Vienna

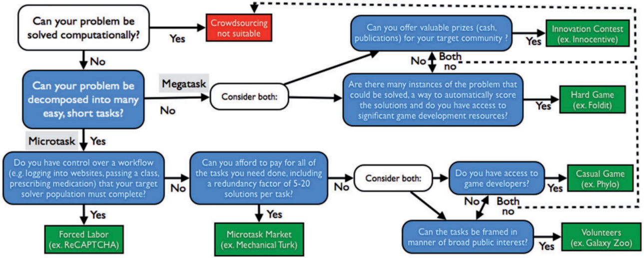
Slide source: [Jia et al., 2015]

2021: AlphaFold is a new breakthrough in protein structure prediction



Source: DeepMind

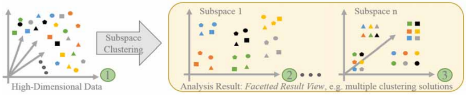
Slide source: [Jumper et al., 2021]



Source: Andreas Holzinger, University of Natural Resources and Life Sciences, Vienna

Slide source: [Good and Su, 2013]

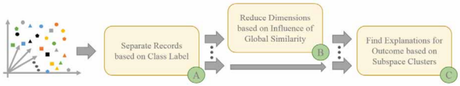
- Patterns may be found in subspaces (dimension combinations)
- Clustering and subset selection: non-convex and NP-hard
- Real data are often noisy and corrupted
- Little prior knowledge about low-dimensional structures
- Data points in different subgroups can be very close



Source: Andreas Holzinger, University of Natural Resources and Life Sciences, Vienna

Slide source: [Hund et al., 2015]

What did the human in the loop do?



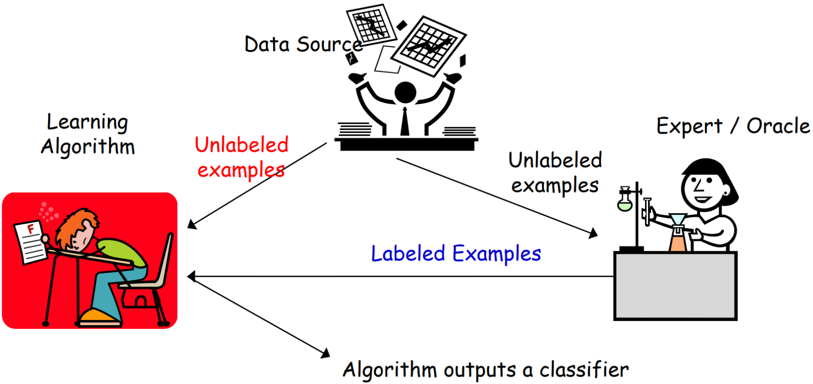
Source: Andreas Holzinger, University of Natural Resources and Life Sciences, Vienna

- **Positive** subspace clusters
  - One homogeneous cluster (healthy patients)
- **Negative** subspace clusters
  - Cluster with obvious reasons for neg. outcome

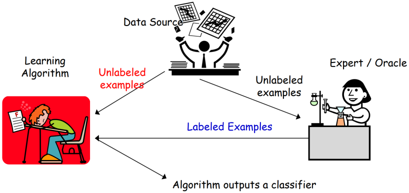
Slide source: [Hund et al., 2015]

Interactive Machine Learning

Semi-supervised Learning



Source: Maria-Florina Balcan, CMU

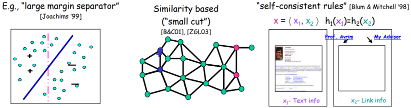


Source: Maria-Florina Balcan, CMU

- $S_l = (x_1, y_1), (x_2, y_2), ..., (x_m, y_m)$ where x_i drawn i.i.d. from D, $y_i = c^*(x_i)$
- $S_u = x_1, ..., x_{mu}$ drawn from i.i.d. D
- Goal: find a classifier with a small error over D
- $err_D(h) = P(h(x) \neq c^*(x))$

Semi-supervised Learning – Key Insights

- Unlabelled data useful if we have a bias/belief not only about the form of the target, but also about its relationship with the underlying data distribution



Source: Maria-Florina Balcan, CMU

- Unlabelled data can help reduce search space or re-order the functions in the search space according to our belief, biasing the search towards functions satisfying the belief (which becomes concrete once we see unlabelled data)

Analyse fundamental sample complexity aspects:

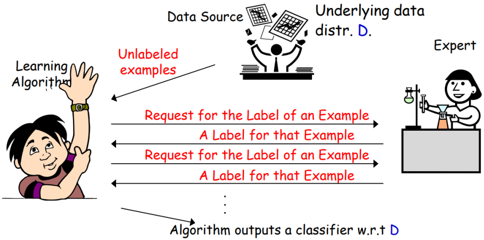
- How much unlabelled data is needed
    - Depends both on complexity of H and of compatibility notion
- Ability of unlabelled data to reduce the number of labelled examples
    - Compatibility of the target, helpfulness of the distribution

Zhu and Goldberg [2009] explain SSL techniques from this point of view

Interactive Machine Learning

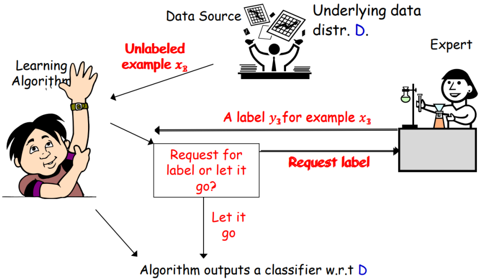
Active Learning

- Two faces of active learning [Dasgupta, 2011]
- Active Learning by Burr Settles [Settles, 2009]
- Active Learning – Modern Learning Theory [Balcan and Urner, 2015]



Source: Maria-Florina Balcan, CMU

- Learner can choose specific examples to be labelled
- Goal: use fewer labelled examples / Pick informative examples to be labelled



Source: Maria-Florina Balcan, CMU

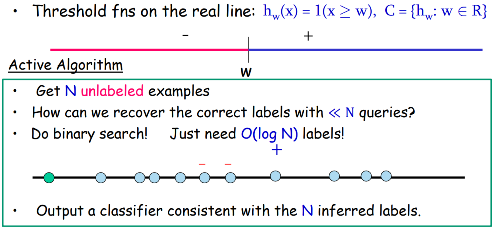
- Selective sampling AL (Online AL): stream of unlabelled examples, when each arrives make a decision to ask for label
- Goal: use fewer labelled examples / Pick informative examples to be labelled

- Guaranteed to output a relatively good classifier for most learning problems
- Does not make too many label requests
  - Hopefully a lot less than passive learning and SSL
- Need to choose the label requests carefully, to get informative labels

Can adaptive querying really do better than passive/random sampling?

- Yes! (sometimes)
- We often need far fewer labels for active learning than for passive
- This is predicted by theory and has been observed in practice

Can it help?



Source: Maria-Florina Balcan, CMU

- $N = O(1/\epsilon)$ we are guaranteed to get a classifier of error $\leq \epsilon$
- **Passive supervised**: $\Omega(1/\epsilon)$ labels to find an ϵ-accurate threshold
- **Active**: only $O(log 1/\epsilon) \rightarrow$ exponential improvement

- Uncertainty sampling in SVMs common and quite useful in practice [Tong and Koller, 2001; Jain et al., 2010; Schohn and Cohn, 2000]

Active SVM Algorithm:

- At any time during the algorithm, we have a "current guess" *wt* of the separator: the max-margin separator of all labelled points so far

- Request the label of the example closest to the current separator

- Active SVM seems to be quite useful in practice [Tong and Koller, 2001; Jain et al., 2010]

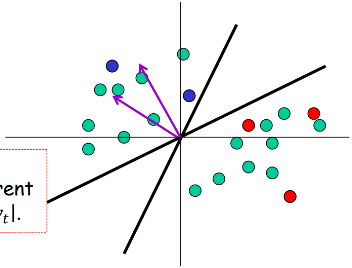
Algorithm (batch version):

- Input $S_u = x_1, \ldots, x_{m_u}$ drawn i.i.d from the underlying source D
- Start: query for the labels of a few random x_is

For $t = 1, \ldots,$

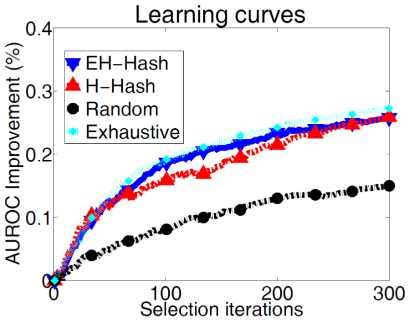
- Find w_t the max-margin separator of all labeled points so far.
- Request the label of the example closest to the current separator: minimizing $|x_i \cdot w_t|$.

(highest uncertainty)



Source: Maria-Florina Balcan, CMU

- Active SVM seems to be quite useful in practice
- E.g. [Jain et al., 2010]



Source: Maria-Florina Balcan, CMU

Newsgroups dataset (20.000 documents from 20 categories)

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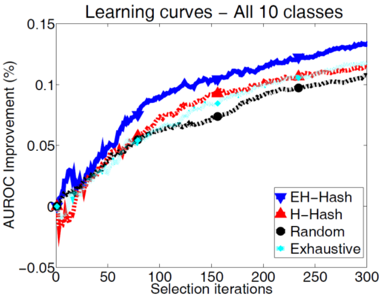
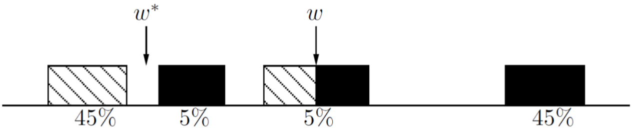


Source: Maria-Florina Balcan, CMU

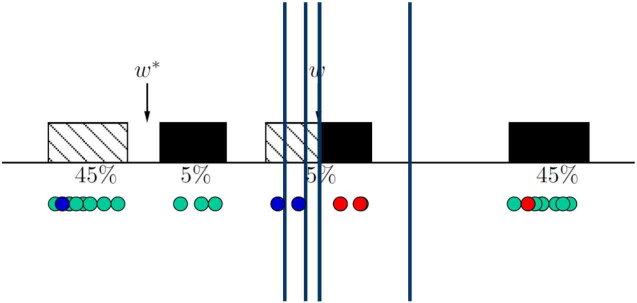
CIFAR-10 image dataset (60.000 images from 10 categories)

- Works sometimes
- **However, we need to be very careful!**
  - Myopic, greedy technique can suffer from **sampling bias**
  - A bias created because of the querying strategy; as time goes on the sample is less and less representative of the true data source



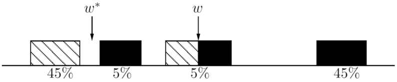
Source: [Dasgupta, 2011]

- Works sometimes
- However, we need to be very careful!



Source: [Dasgupta, 2011]

- Works sometimes
- **However, we need to be very careful!**
  - Myopic, greedy technique can suffer from **sampling bias**
  - A bias created because of the querying strategy; as time goes on the sample is less and less representative of the true data source
  - Observed in practice too!



Source: Maria-Florina Balcan, CMU

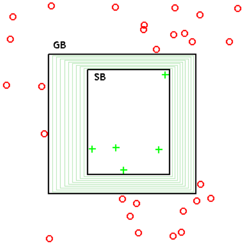
- Main tension: want informative points, but also want guarantees that the classifier does well on true random examples from the underlying distribution

- Safe Active Learning Schemes
- Disagreement Based Active Learning
- Hypothesis Space Search

- X – feature/instance space; distr. D over X; $c^* \in H$ target fnc
- Fix hypothesis space H

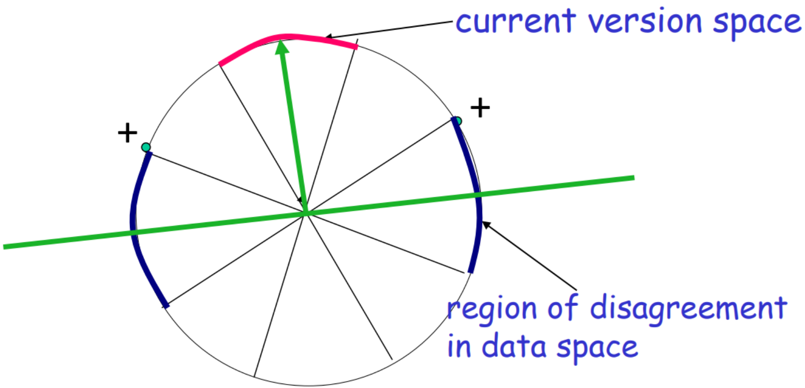
Definition [Mitchell, 1982]:

- Assume realisable case: $c^* \in H$
- Given a set of labelled examples $(x_1, y_1), \ldots, (x_m, y_m)$, $y_i = c^*(x_i)$
- **Version space of H**: part of H that is consistent with the labels so far
- i.e. $h \in VS(H)$ iff $h(x_i) = c^*(x_i)\ \forall i \in \{1, \ldots, m_i\}$



Source: *https://en.wikipedia.org/wiki/Version_space_learning*

- GB is the maximally general positive hypothesis boundary
- SB is the maximally specific positive hypothesis boundary
- The intermediate (thin) rectangles represent the hypotheses in the version space



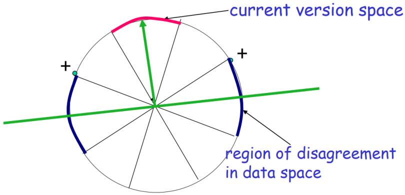
Source: Maria-Florina Balcan, CMU

E.g.: data lies on circle in R^2, H = homogeneous linear separators

Definition [Cohn et al., 1992]:

- Version space: part of H consistent with the labels so far
- Region of disagreement: part of data space about which there is still some uncertainty (i.e. disagreement within version space)

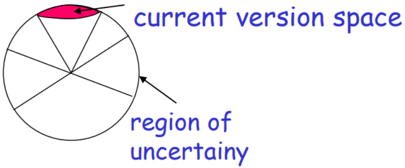
$$x \in X, x \in DIS(VS(H)) iff \exists h_1, h_2 \in VS(H), h_1(x) \neq h_2(x)$$



Source: Maria-Florina Balcan, CMU

Algorithm:

- Pick a few points at random from the current region of uncertainty and query their labels
- Stop when region of uncertainty is small

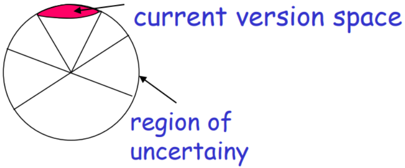


Source: Maria-Florina Balcan, CMU

Why is it active learning? We do not waste labels by querying in regions of space we are certain about the labels

Algorithm:

- Query for the labels of a few random x_is
- Let H_t be the current version space
- for $t = 1 \ldots$
  - Pick a few points at random from the current region of disagreement $DIS(H_t)$ and query their labels
  - Update the new version space to H_{t+1}



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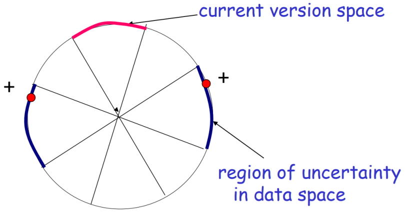
- Current version space: part of C consistent with labels so far
- “Region of uncertainty” = part of data space about which there is still some uncertainty (i.e. disagreement within version space)



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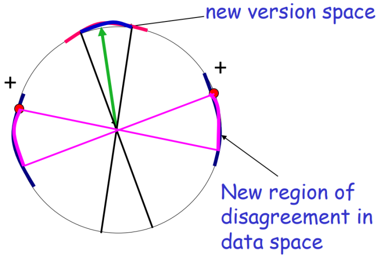


Source: Maria-Florina Balcan, CMU

How about the agnostic case where the target might not belong the H?

A^2 algorithm [Balcan et al., 2006]:

- Let $H_1 = H$
- for $t = 1 \ldots$
  - Pick a few points at random from the current region of disagreement $DIS(H_t)$ and query their labels
  - Throw out hypothesis if you are statistically confident they are suboptimal

Note: Careful use of generalization bounds; Avoid the sampling bias!

- A^2 is the first algorithm robust to noise [Balcan et al., 2006, 2009]:
- "Region of disagreement" style: Pick a few points at random from the current region of disagreement, query their labels, throw out hypothesis if you are statistically confident they are suboptimal

(theoretical) Guarantees:

- It is safe (never worse than passive learning) & exponential improvements
- C – thresholds, low noise, exponential improvement
- C - homogeneous linear separators in R^d, D - uniform, low noise, only $d^2 log(1/\epsilon)$ labels

How quickly the region of disagreement collapses as we get closer and closer to optimal classifier

Guarantees for A^2 [Hanneke'07]:

Disagreement coefficient $\theta_{c^*} = \sup_{r \geq \eta+\epsilon} \frac{\Pr(DIS(B(c^*, r)))}{r}$

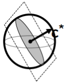
Theorem

$$m = \left(1 + \frac{\eta^2}{\epsilon^2}\right) VCdim(C)\theta_{c^*}^2 \log\left(\frac{1}{\varepsilon}\right)$$

labels are sufficient s.t. with prob. $\geq 1-\delta$ output h with $err(h) \leq \eta+\epsilon$.

Realizable case: $m = VCdim(C)\theta_{c^*} \log\left(\frac{1}{\varepsilon}\right)$

Linear Separators, uniform distr.: $\theta_{c^*} = \sqrt{d}$



Source: Maria-Florina Balcan, CMU

Further reading: [Hanneke, 2007]

"Disagreement based" algorithms: query points from current region of disagreement, throw out hypotheses when statistically confident they are suboptimal

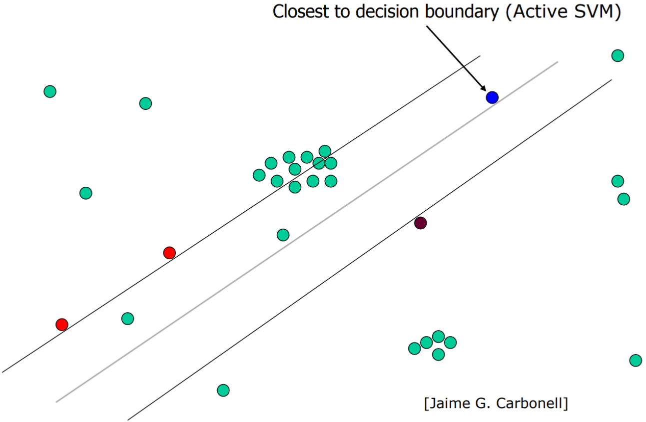
- Generic (any class), adversarial label noise
- Computationally efficient for classes of small VC-dimension (Vapnik–Chervonenkis dimension)

Still, could be suboptimal in label complexity and computationally inefficient in general

Interactive Machine Learning

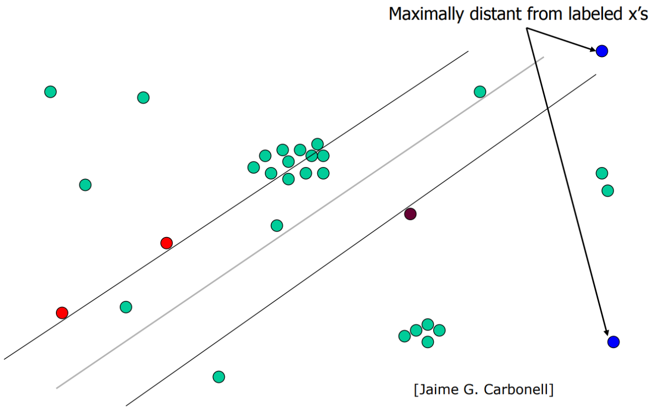
Other Interesting AL Techniques used in Practice

Uncertainty Sampling

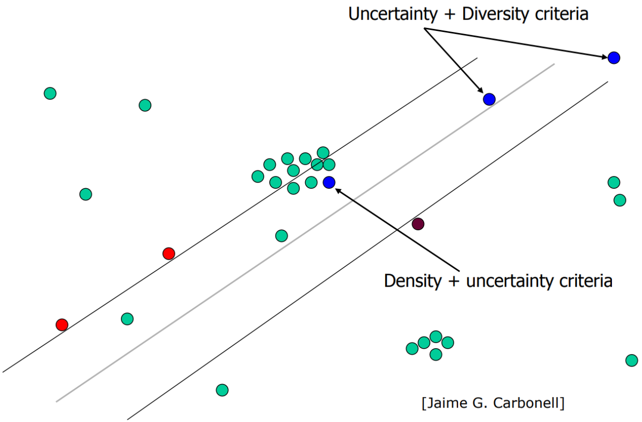


Source: Maria-Florina Balcan, CMU

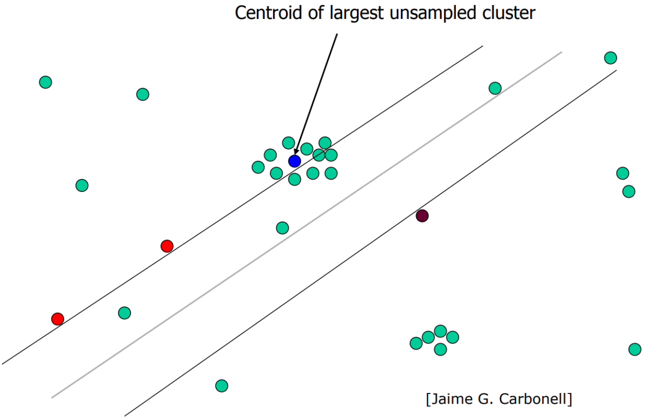
Maximal Diversity Sampling



Source: Maria-Florina Balcan, CMU



Source: Maria-Florina Balcan, CMU

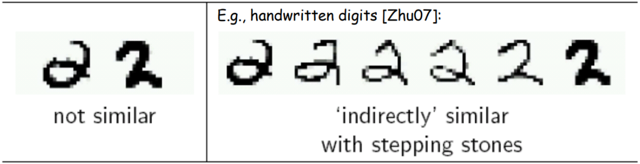


Source: Maria-Florina Balcan, CMU

Interactive Machine Learning

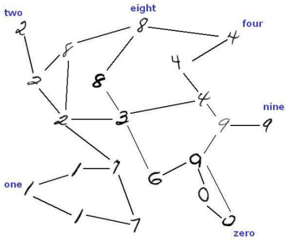
Graph-based Active and Semi-supervised Methods

- Assume we are given a pairwise similarity function and that very similar examples probably have the same label
- If we have a lot of labelled data, this suggests a Nearest-Neighbour type of algorithm
- If you have a lot of unlabelled data, perhaps can use them as "stepping stones"



Source: Maria-Florina Balcan, CMU

- Key idea: construct a graph with edges between very similar examples
- Unlabelled data can help “glue” the objects of the same class together

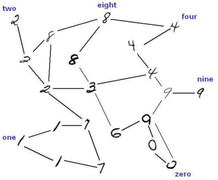


Source: Maria-Florina Balcan, CMU

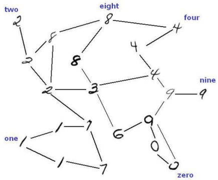
Often, transductive approach: Given L + U, output predictions on U. Are allowed to output any labelling of $L \cup U$

Main Idea:

- Construct graph G with edges between very similar examples
- Might have also glued together in G examples of different classes
- Run a graph partitioning algorithm to separate the graph into pieces



Source: Maria-Florina Balcan, CMU



Source: Maria-Florina Balcan, CMU

Graph partitioning algorithms:

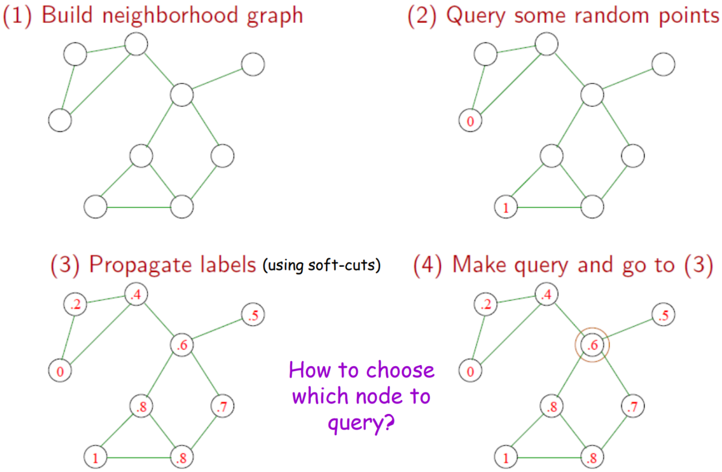
- Minimum/Multiway cut [Blum and Chawla, 2001]
- Minimum "soft-cut" [Zhu et al., 2003]
- Spectral partitioning

[Zhu et al., 2003]

Solve for label function $f(x) \in [0, 1]$ to minimise:

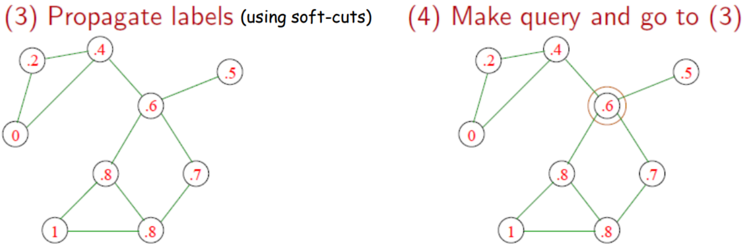
$$J(f) = \underbrace{\sum_{edges\,(i,j)} w_{ij}\left(f(x_i) - f(x_j)\right)^2}_{\text{Similar nodes get similar labels (weighted similarity)}} + \underbrace{\sum_{x_i \in L} \lambda(f(x_i) - y_i)^2}_{\text{Agreement with labels (agreement not strictly enforces)}}$$

Source: Maria-Florina Balcan, CMU



Source: Maria-Florina Balcan, CMU

- One natural idea: query the most uncertain point
- But it can have only one edge. Query will not have much impact!



Source: Maria-Florina Balcan, CMU

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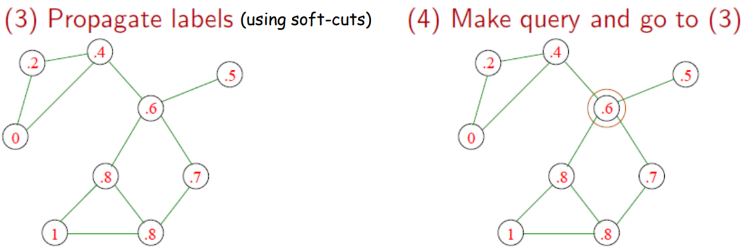


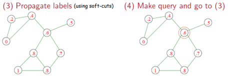
Source: Maria-Florina Balcan, CMU

Instead use 1-step-lookahead heuristic:

- For a node with label p, assume that querying will have prob p of returning answer 1, $1-p$ of returning 0
- Compute "average confidence" after running soft-cut in each case

$$p\frac{1}{n}\sum_{x_i} max(f_1(x_i), 1-f_1(x_i)) + (1-p)\frac{1}{n}\sum_{x_i} max(f_0(x_i), 1-f_0(x_i))$$

- Query node s.t. this quantity is highest (you want to be more confident on average)



Source: Maria-Florina Balcan, CMU

Does well for Video Segmentation [Fathi et al., 2011]

Source: Maria-Florina Balcan, CMU

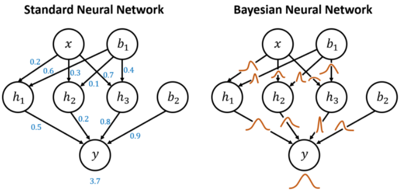
- Active learning can be really helpful!
- Can provide exponential improvements in label complexity, both theoretically and practically
- Common heuristics (e.g. those based on uncertainty sampling) – careful for sampling bias!
- Safe Disagreement Based Active Learning Schemes
  - Understand how they operate precisely in noise-free scenarios

Interactive Machine Learning

Deep Active Learning

- Challenging, ongoing research!
- Deep neural networks usually lack measures of uncertainty.
  - Output of final softmax layer tends to be overconfident
- Single image vs. batch of image for large-scale optimisation
  - How should we select a batch of images at once?

- Bayesian neural networks give better uncertainty measures
- In a Bayesian neural network, every parameter in the model is sampled from a distribution



Source: cyda: `https://blog.cyda.hk/`

Idea: An uncertainty measure from a single network might be flawed – go over many networks to improve that

Slide source: [Gal et al., 2017]

Monte Carlo dropout

- Intractable to integrate over all possible parameter values in the distribution

$$p(y=c|x) = \int p(y=c|x,\omega)p(\omega)d\omega$$

$$\sim \int p(y=c|x,\omega) * q^*(\omega)d\omega$$

$$\sim \frac{1}{T}\sum_t p(y|x,\omega_t) = \frac{1}{T}\sum_t p_c^t$$

We do not know the actual real parameter distribution $\rightarrow$ approximate using a Bernoulli distribution

Slide source: [Gal et al., 2017]

$$H \sim -\sum_c \left(\frac{1}{T}\sum_t p_c^t\right) Log\left(\frac{1}{T}\sum_t p_c^t\right)$$

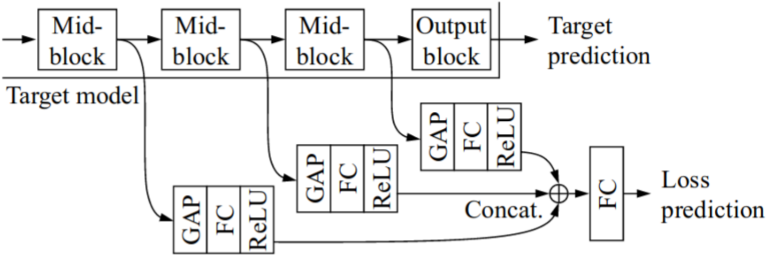
- We need to run the network multiple times with dropout, average the outputs, and take the entropy
- High H:
  - Every time we run the model with dropout, it is confident about a different category. **If we select this image, we can correct those parameters that caused the models to be wrong**
  - Models agree about the input X but are not so confident

- Find examples for which many of the different sampled networks are wrong about
- Find the image that maximises the mutual information between the model output and the model parameters

$$I(y;\omega|x, D_{train}) = H(y|x, D_{train}) - E_{p(\omega, D_{train})}[H(y|x, \omega, D_{train})]$$

- The first term looks for images that have high entropy in the average output
- The second term penalizes images where many of the sampled models are not confident about, i.e. **disagree** on

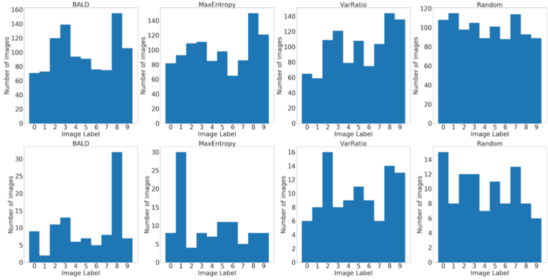
Key idea: try to predict the learning loss of an algorithm



Source: [Yoo and Kweon, 2019]

- Extract features from intermediate layers and combine them to predict the network loss
- Problem? Loss changes over time…

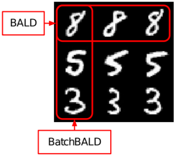
- Extract features from intermediate layers and combine them to predict the network loss
- Problem? Loss changes over time...
- Comparing the predicted losses between images
- Every batch of size B is split into B/2 pairs of images
- In every pair, the loss is predicted for both images

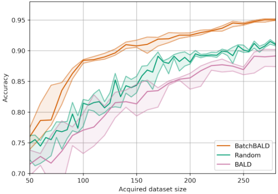


Source: [Pop and Fulop, 2018]

MNIST histograms of true labels in the training set. Top: End of AL process. Total number of images in training set: 1,000. Bottom: After the first 8 acquisition iterations. Total number of images in the training set: 100

Uncertainty vs Diversity Sampling





Source: [Kirsch et al., 2019]

Problem: Datasets can have many near duplicates. If we just select the top-ranking images, we might select many near-duplicate images that all rank high

- A Survey of Deep Active Learning [Ren et al., 2022]
- A Survey of Active Learning for Text Classification using Deep Neural Networks [Schröder and Niekler, 2020]
- A Survey on Active Learning and Human-in-the-Loop Deep Learning for Medical Image Analysis [Budd et al., 2021]

Generative Models I

- Mixture of Gaussians
- Autoencoders
- Variational Autoencoders (VAEs)

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